PROGRAM:

```
clc;
clear;
close all;
% Parameters
fs = 10000; % Sampling frequency (Hz)
Am = 1; % Message amplitude
Ac = 1; % Carrier amplitude
fm = 35; % Message frequency (Hz)
fc = 500; % Carrier frequency (Hz)
B = 10; % Modulation index
freqdev = B * fm; % Frequency deviation (Δf = B * fm)
kf = 2 * pi * freqdev; % Frequency sensitivity
t = (0:0.1*fs)/fs; % Time vector (0 to 0.1 seconds)
% Message signal
m_t = Am * cos(2 * pi * fm * t);
% Carrier signal (for reference)
c_t = Ac * cos(2 * pi * fc * t);
% === FM Modulation (manual) ===
int_m = cumsum(m_t) / fs; % Integral of message
s_t = Ac * cos(2*pi*fc*t + kf * int_m); % FM signal
%% ---------------- Figure 1: Modulation ----------------
figure(1);
subplot(3,1,1);
plot(t, m_t, 'b');
title('Message Signal (fm = 35 Hz) SAMIN');
xlabel('Time (s)'); ylabel('Amplitude');
grid on;
subplot(3,1,2);
plot(t, c_t, 'g');
title('Carrier Signal (fc = 500 Hz) SAMIN');
xlabel('Time (s)'); ylabel('Amplitude');
grid on;
subplot(3,1,3);
plot(t, s_t, 'r');
title('FM Modulated Signal (B = 10) SAMIN');
xlabel('Time (s)'); ylabel('Amplitude');
grid on;
%% ========= FM Demodulation (manual, using phase) =========% Hilbert
Transform to get analytic signal
s_analytic = hilbert(s_t); % Complex envelope
inst_phase = unwrap(angle(s_analytic)); % Instantaneous phase
inst_freq = [0 diff(inst_phase)] * fs / (2*pi); % Instantaneous freq (Hz)
% Remove carrier frequency to get message
m_demod = (inst_freq - fc) / freqdev; % Normalized back to message
%% ---------------- Figure 2: Demodulation ----------------
figure(2);
subplot(2,1,1);
plot(t, m_t, 'b--');
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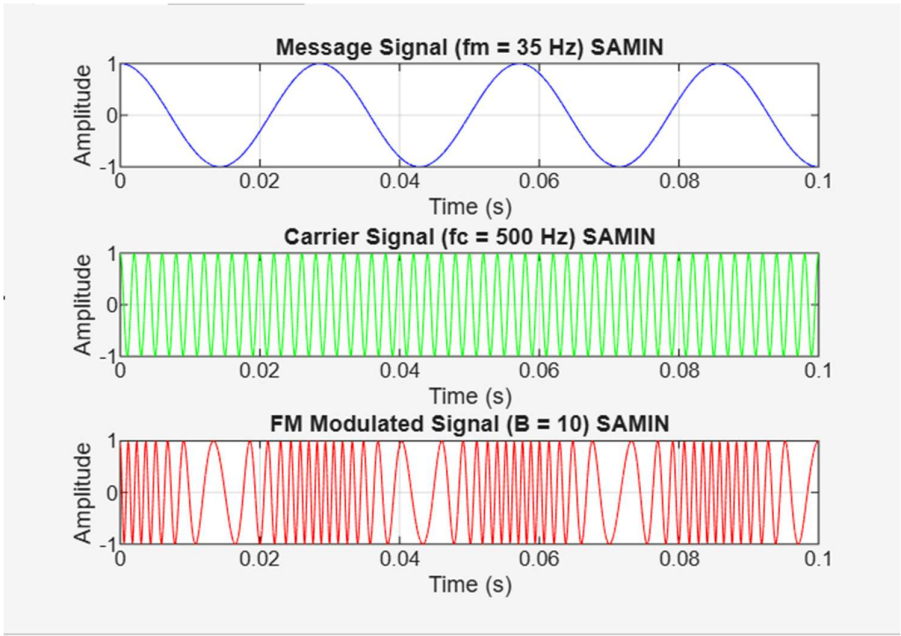
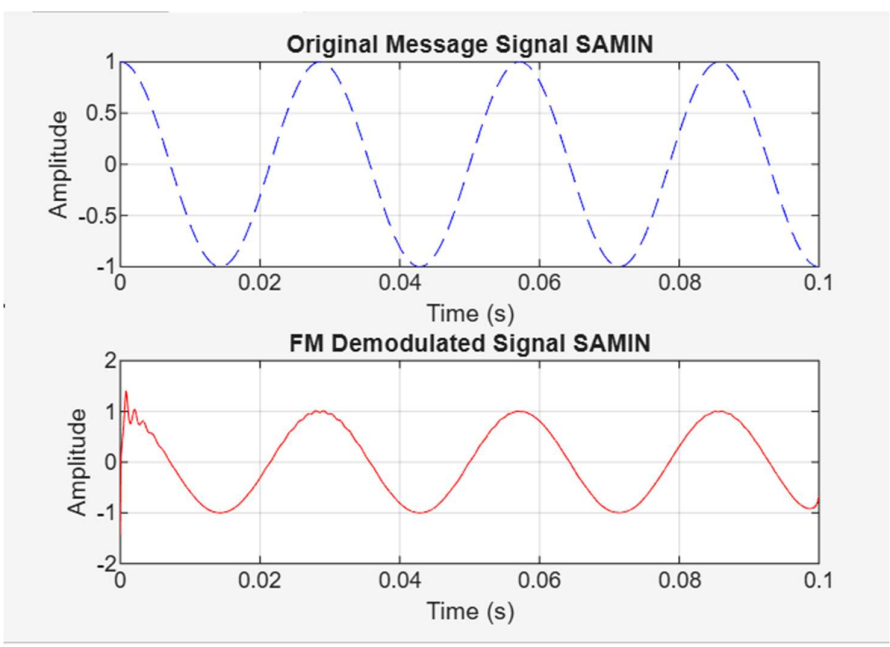
```
title('Original Message Signal SAMIN');
xlabel('Time (s)'); ylabel('Amplitude');
grid on;
subplot(2,1,2);
plot(t, m_demod, 'r');
title('FM Demodulated Signal SAMIN');
xlabel('Time (s)'); ylabel('Amplitude');
grid on;
```

OUTPUT: